ModelIt!

Student Activity Pack

Why Do Athletes Collapse in the Heat?

Modeling Homeostasis and the Body's Breaking Point

Grade 09L1 | NGSS HS-LS1-3 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think some athletes collapse from the heat while others on the same team are fine?

2 What happens inside your body when you get too hot during exercise?

3 Draw a diagram showing how you think the body cools itself during intense physical activity.

4 What is homeostasis and why is it important for survival?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Core Temperature | Sweat Rate | Hydration Level | Physical Exertion

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Moderate & Hydrated

Moderate exertion, high hydration

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Intense & Dehydrated

Maximum exertion, low hydration

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Gradual Dehydration

Steady exertion, declining hydration

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Ambient Temperature — The external air temperature and humidity, which affects how efficiently sweat can evaporate and cool the body

Blood Volume — The total volume of blood circulating in the body, which decreases with dehydration and reduces the body's ability to transport heat

Acclimatization Level — How adapted an athlete's body is to exercising in heat, which improves sweating efficiency and reduces strain

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do some athletes collapse during games while others playing the same sport are fine? Use evidence from your simulation to support your answers.

1 How does your model demonstrate that homeostasis depends on negative feedback loops?

2 What happens in your model when the feedback loop is broken by dehydration?

3 Why might two athletes of the same fitness level respond differently to the same heat conditions?

4 How could coaches use your model to design safer practice schedules?

5 What are the limitations of modeling a complex biological system with only four components?

8 STEM Challenge: Design a Heat Safety Alert System

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YOUR ENGINEERING MISSION

Design a wearable or sideline monitoring system that tracks athlete biometrics and alerts coaches before heat-related illness occurs.

THE SCENARIO

A high school athletic department has had three heat-related incidents in two years. The principal has hired your team to design a monitoring system that can predict when an athlete is approaching dangerous territory and alert the coaching staff before collapse occurs.

Engineering Design Thinking

1. WHAT SENSORS WOULD YOU NEED AND WHERE WOULD THEY BE PLACED ON THE ATHLETE?

2. HOW WOULD THE SYSTEM COMMUNICATE ALERTS TO COACHES DURING A GAME?

3. HOW WOULD YOU ACCOUNT FOR DIFFERENCES IN BODY SIZE, FITNESS LEVEL, AND ACCLIMATIZATION?

4. WHAT DATA WOULD YOU DISPLAY ON THE COACH'S DASHBOARD?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Sports Physiologists study how the body responds to physical activity and environmental stress. They work with professional sports teams, military programs, and research institutions, earning $60,000–

$120,000/year.